Data Genie — Project Proposal

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1. Problem Statement / Proposal

Data Genie is a **command-line database management and analytics application** that lets users interact with SQLite databases through simple natural-language-style requests. Many people find databases difficult to use because they must learn SQL commands, table structures, and database terminology. This project will make common database operations easier by translating supported user requests into SQL commands.

Users will be able to connect a SQLite database, inspect its tables and columns, create tables, add records, search data, update records, delete records, and request basic analysis. For example, a user could ask to "show all customers from Missouri," "add a new employee," or "calculate the average sale amount by category." Data Genie will generate the needed SQL, execute it, and display the results as text tables or summaries. Destructive commands, such as deleting records or dropping tables, will require user confirmation.

SQLite will be the only supported database system for the first version to keep the project realistic for one semester. The project may use predefined command patterns for interpreting requests; more advanced LLM-based interpretation or a web interface may be considered only if time permits.

A database system is essential because Data Genie's main purpose is to store, organize, retrieve, modify, and analyze structured information. The database contains the tables, records, relationships, and schema that the application uses to answer requests and perform management operations.

2. Functional Requirements

1. **Connect to a SQLite database** — Users can select a SQLite database file for Data Genie to use.
   Entities: Data Source, Database, Table, Column
2. **Inspect database structure** — Users can view database tables, columns, data types, and schema details.
   Entities: Database, Table, Column
3. **Create tables** — Users can create new tables and define columns through supported natural-language-style commands.
   Entities: Database, Table, Column, SQL Command
4. **Insert records** — Users can add new records to a selected table.
   Entities: Table, Column, Record, SQL Command
5. **Search and retrieve data** — Users can query data using filters, selected columns, sorting, grouping, and basic joins.
   Entities: Database, Table, Column, Record, Query, SQL Command
6. **Update and delete records** — Users can change or remove existing records. Data Genie will request confirmation before destructive delete operations.
   Entities: Table, Column, Record, SQL Command

7. **Modify or delete tables** — Users can add columns or remove tables when supported by SQLite. The system will confirm before running `DROP TABLE`.
   Entities: Database, Table, Column, SQL Command
8. **Analyze data** — Users can request totals, averages, counts, category comparisons, groupings, and basic trend summaries.
   Entities: Database, Table, Column, Record, Query, Analysis Result
9. **Display results** — The program will show query results and analysis results in readable text tables and summaries.
   Entities: Query, Record, Analysis Result
10. **Store command history** — The system will keep a session history of user requests, generated SQL commands, and results.
    Entities: User Request, Query, SQL Command, Result